

MUKESH GHIMIRE

mukeshghimire.com.np | mghimire@asu.edu

in |  |  | +1-662-202-2139

SUMMARY

Research scientist working at the intersection of reinforcement learning, LLM post-training, and agent evaluation. Develops learning algorithms and empirical methodologies for language models and autonomous systems, with research spanning unverifiable rewards, game-theoretic decision-making, and safe multi-agent interaction.

RELEVANT EXPERIENCE

Amazon Web Services (AWS) - Santa Clara, CA

Applied Scientist II, 06/2026 - present

- Research and develop evaluation methods for autonomous agents and tool-using language models, measuring task success, robustness to edge cases, and behavioral consistency.
- Design evaluation pipelines based on agent execution traces, automated model-based judging, custom evaluators, and offline/online assessment to diagnose and improve agent behavior.
- Investigate evaluation signals and reward formulations for agentic systems, connecting empirical measurements to post-training and reinforcement-learning workflows.
- Collaborate across research and engineering teams to translate advances in agent evaluation into scalable systems for real-world deployment.

Mercedes-Benz R&D - San Jose, CA

Machine Learning Intern, 09/2025 - 01/2026

- Built a vision-language-action model family using Qwen as a backbone, replacing discrete token generation with a continuous action policy.
- Pre-trained the model on nuScenes to predict future ego-trajectories from multimodal inputs, including camera observations and historical vehicle state.

Arizona State University - Tempe, AZ

Ph.D. Researcher, 06/2021 - 03/2026

- Developed scalable algorithms for one-sided incomplete-information differential games, deriving explainable and safety-aware strategies for autonomous systems.
- Formulated autonomous-vehicle and multi-agent interactions as general-sum and zero-sum differential games and computed equilibrial policies for vehicles and swarms.
- Designed an RL-based controller that reduced the frequency of intent inference in human-autonomous-agent interactions while preserving decision quality.
- Published research at ICML, ICLR, and ACC, and reviewed papers for ICLR, ICML, NeurIPS, and AAAI.

SKILLS

Reinforcement Learning: RLHF, RLVR, policy optimization, reward modeling, multi-agent RL, exploration, safe RL

LLM Research: LLM post-training, agent evaluation, LLM-as-a-Judge, test-time learning, vision-language-action models, model collapse mitigation

Mathematical & Theoretical: Differential Games, Game Theory, Stochastic Processes, Optimal Control, Numerical Methods for PDEs

Technical: Python (JAX, NumPy, PyTorch), Linux, Git

EDUCATION

Arizona State University

Tempe, AZ, USA

Degree: Ph.D.

Dissertation: Exploiting Equilibrium Structure for Scalable Solutions to Zero-Sum Differential Games with One-Sided Information

Advisors: Yi Ren, Zhe Xu

08/2021 - 03/2026

SELECTED PUBLICATIONS

- [1] **M. Ghimire**, A. Feng, L. You, Y. Luo, F. Liu, X. Zhu. *Stabilizing Post-Training without Task-Verifiable Rewards via Process Rewards and Internal Confidence*. Under Review.
- [2] **M. Ghimire**, L. Zhang, Z. Xu, Y. Ren. *Solving Football by Exploiting Equilibrium Structure of 2p0s Differential Games with One-Sided Information*. Accepted at ICLR'26.
- [3] **M. Ghimire**, L. Zhang, Z. Xu, Y. Ren. *State-Constrained Zero-Sum Differential Games with One-Sided Information*. ICML'24.
- [4] **M. Ghimire**, L. Zhang, W. Zhang, Y. Ren, Z. Xu. *Solving Two-Player General-Sum Games Between Swarms*. ACC'24.

Full list on Google Scholar.

SELECTED AWARDS & HONORS

Best Paper Award in MARW Workshop at AAAI 2025

Experiential Learning Grant 2023, 2024

GPSA Travel Grant Award 2023

ICRA Travel Grant 2023

SMBHC Research Fund Award 2020

ACADEMIC ACTIVITIES

TALKS: Sparky's Cup Education - Lightning Talk on Game-Changing AI Applications in Sport.

REVIEWING: AAAI (Program Committee '27), ICLR, ICML, NeurIPS.